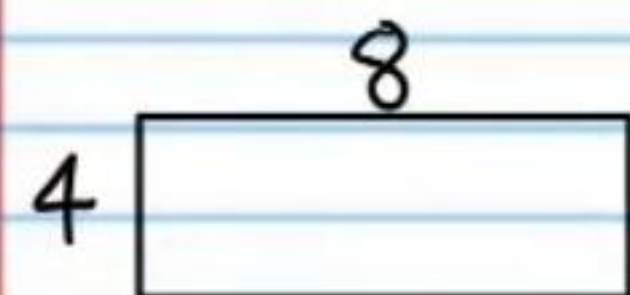


Class 23: The Order of Operations Agreement and Applying the Agreement to Geometric Formulas

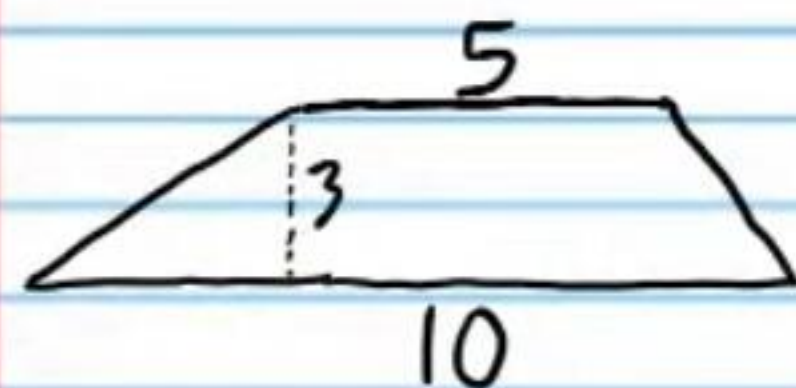
Example I:



$$\begin{aligned} \text{i) } P &= 2 \times 4 + 2 \times 8 \\ &= 2 \times 6 \times 8 \\ &= 12 \times 8 \\ &= \boxed{96} \end{aligned}$$

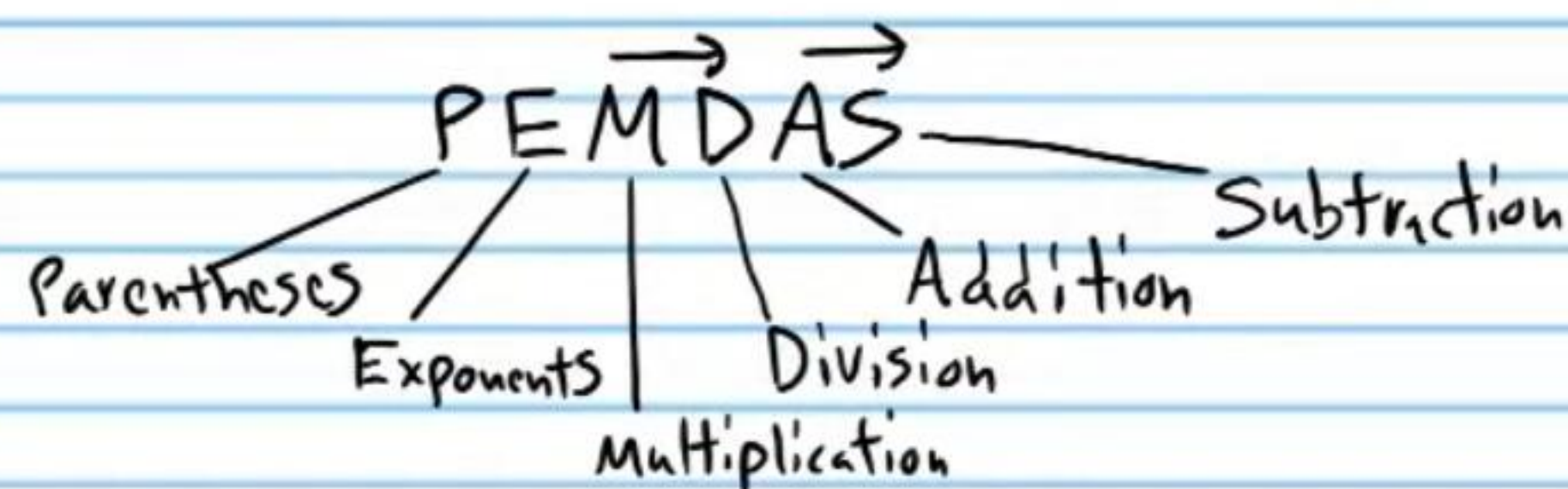
$$\begin{aligned} \text{ii) } P &= 2 \times 4 + 2 \times 8 \\ &= 8 + 16 \\ &= \boxed{24} \end{aligned}$$

Example II:



$$\begin{aligned} \text{i) } A &= 5 + 10 \times 3 \div 2 \\ &= 5 + 30 \div 2 \\ &= 5 + 15 \\ &= \boxed{20} \end{aligned}$$

$$\begin{aligned} \text{ii) } A &= 5 + 10 \times 3 \div 2 \\ &= 15 \times 3 \div 2 \\ &= 45 \div 2 \\ &= \boxed{22.5} \end{aligned}$$



PEMDAS

Perform the Following Operations

① $2 \times 3 + 1$ ② $5 - 9 \div 3$ ③ $2^3 \times 5$

④ $7^2 - 15 \div 3$ ⑤ $63 \div (5 + 4)$ ⑥ $(5 + 3) \times 7$

⑦ $1 + 3 \times (8 - 4)$ ⑧ $72 \div (1 + 2)^2$ ⑨ $30 - (5^2 - 20)^2$

⑩ $(100 \div 10 - 3^2) + 8 \times 2^3$ ⑪ $5 \times 3 + (2 \times 1 + 2^4)$

Why do we multiply and divide left to right and add and subtract left to right?

⑫ i) $8 \div 4 \times 2$ ii) $8 \div 4 \times 2$ ⑬ i) $7 - 3 + 1$ ii) $7 - 3 + 1$

$8 \div 8$ 2×2 $7 - 4$ $4 + 1$

$\boxed{1}$ X $\boxed{4}$ ✓ $\boxed{3}$ X $\boxed{5}$ ✓

⑭ $64 \div 8 \times 5 \div 10$ ⑮ $10 - 2 + 4 - 1$ ⑯ $3 \times 10 \div 5 \times 5$ ⑰ $4 + 9 - 2 + 2$

PEMDAS

⑱ $(62 - 20 \times 2 + 8) \div 10 \times 4$

⑲ $77 \div 11 \times 9 - 4 \times (6 - 3 + 5)$

⑳ $4 \times 9 \div (2^2 - 1 + 6) \times 8 + 1$

㉑ $10 - 3 + 5 \times (3^3 \div 9 \times 2)$

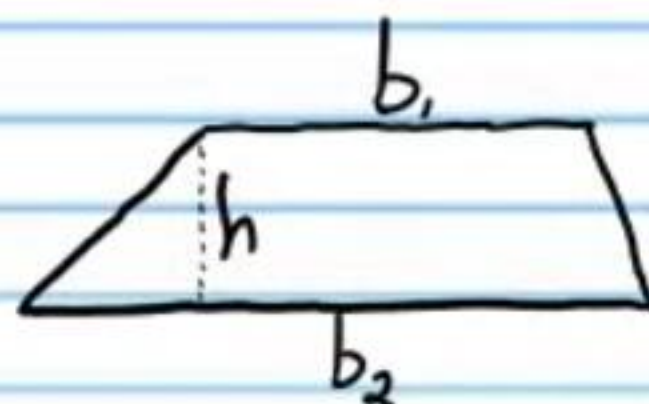
㉒ $[8 - 6 \times (2 - 1)^3] + 7$

㉓ $5 + 72 \div [1 + (5 - 3)^2 + 4]$

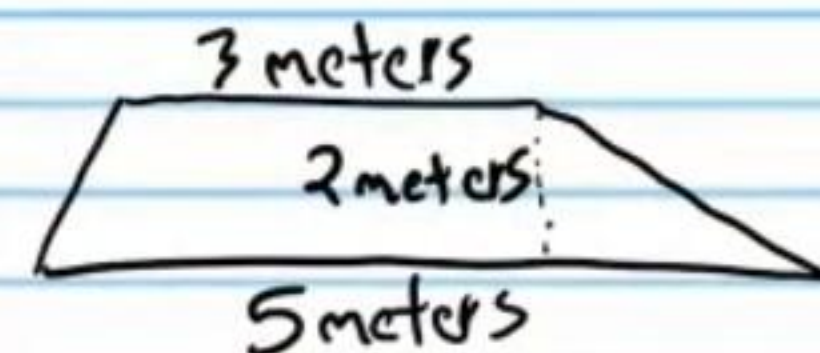
Applying the Order of Operations Agreement to Geometric Formulas

i) Area of a Trapezoid:

$$A = (b_1 + b_2) \times h \div 2$$

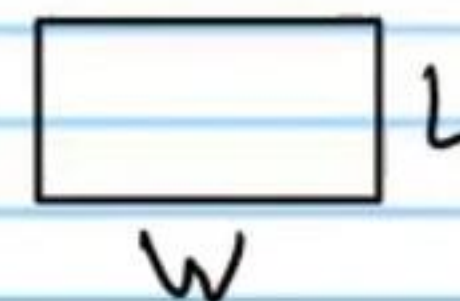


㉔ Find the area of the trapezoid below.

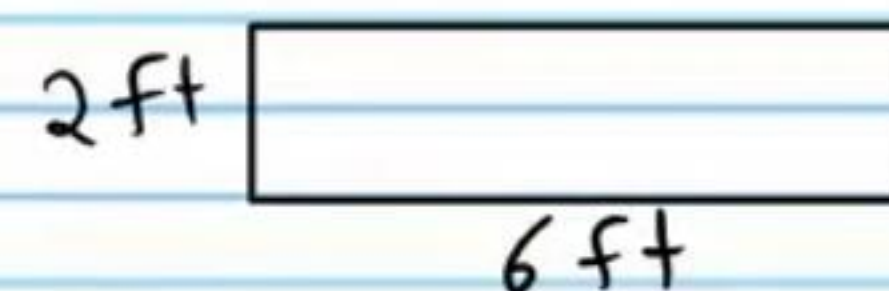


ii) Perimeter of a Rectangle:

$$P = 2 \times W + 2 \times L$$

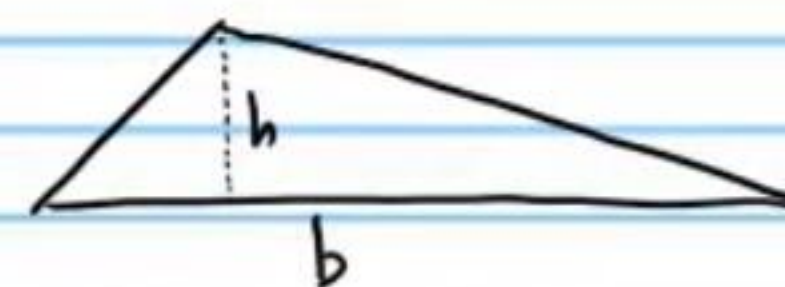


㉕ Find the perimeter of the rectangle below by using the perimeter formula.

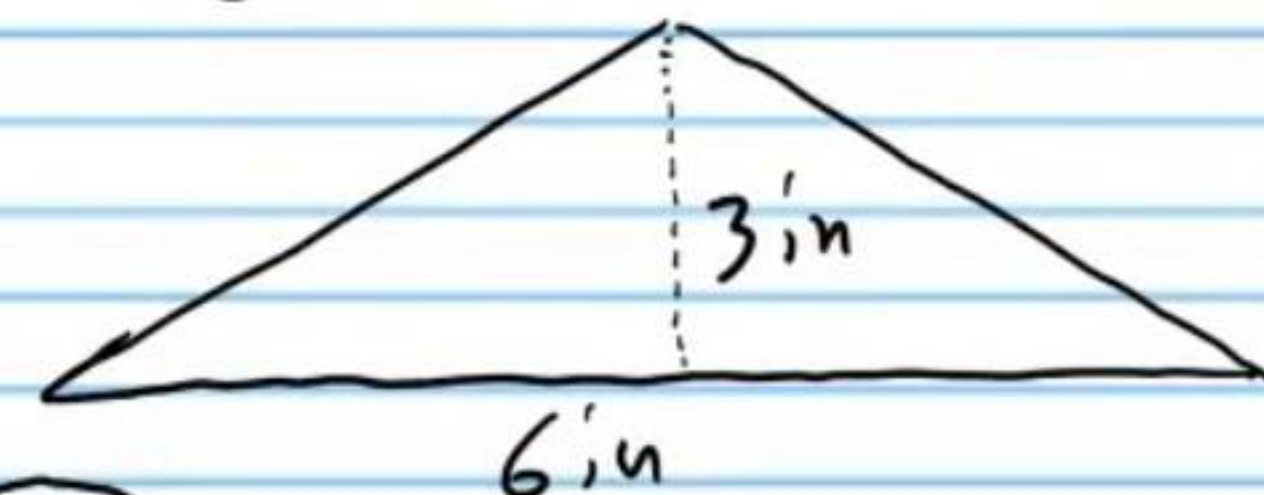


iii) Area of a Triangle:

$$A = b \times h \div 2$$



㉖ Find the area of the triangle below.



iv) Area of a Circle:

$$A \approx 3.14 \times r^2$$



②7 Find the approximate area of a circle with radius four kilometers.

V) Volume of a Sphere:

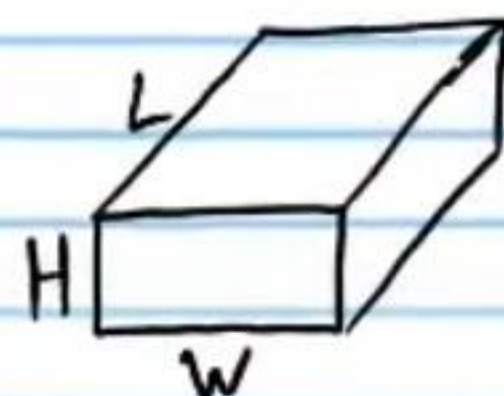
$$V \approx 4 \times 3.14 \times r^3 \div 3$$



Challenge ②8 Approximate the volume of a sphere with radius two centimeters

V) Volume of a Rectangular Prism:

$$V = L \times W \times H$$



②9 Find the volume of a rectangular prism with height 8 ft, length 6 ft, and width 5 ft.

③0 Find the volume of a room in the shape of a rectangular prism with dimensions 1.7 m by 0.89 m by 2 m.

③1 Find the volume of a candy bar in the shape of a rectangular prism with width 3 cm, height 7 cm and length 8 cm.

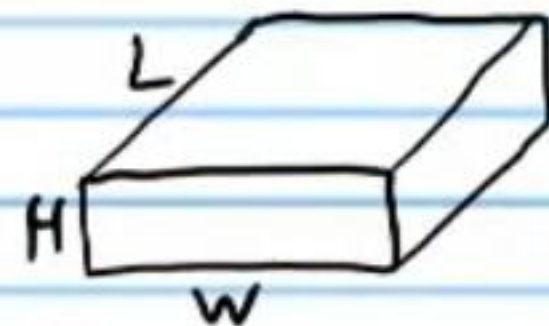
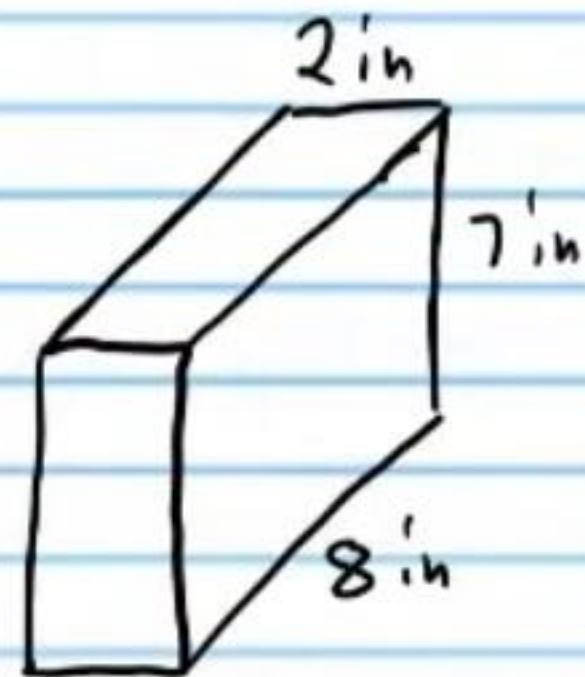
③② Find the volume of a rectangular rod with dimensions 7 in X 8 in X 21 in.

③③ Find the volume of an office building in the shape of a rectangular prism with a length $\frac{1}{7}$ of a kilometer, width $\frac{3}{9}$ of a kilometer, and height $\frac{1}{10}$ of a kilometer.

Vii) Surface Area of a Rectangular Prism:

$$A = 2 \times L \times H + 2 \times W \times L + 2 \times W \times H$$

Example:



③④ Find the surface area of a suitcase in the shape of a rectangular prism with height 11 feet, width 9 feet, and length 3 feet.

③⑤ Find the surface area of a building with dimensions 3 by 10 by 4.

③⑥ Find the surface area of a closed box with dimensions 3 in X 5 in X 9 in.

Viii) Perimeter of a Circle (called circumference):

$$C \approx 2 \times 3.14 \times r \quad \text{OR} \quad C \approx D \times 3.14$$



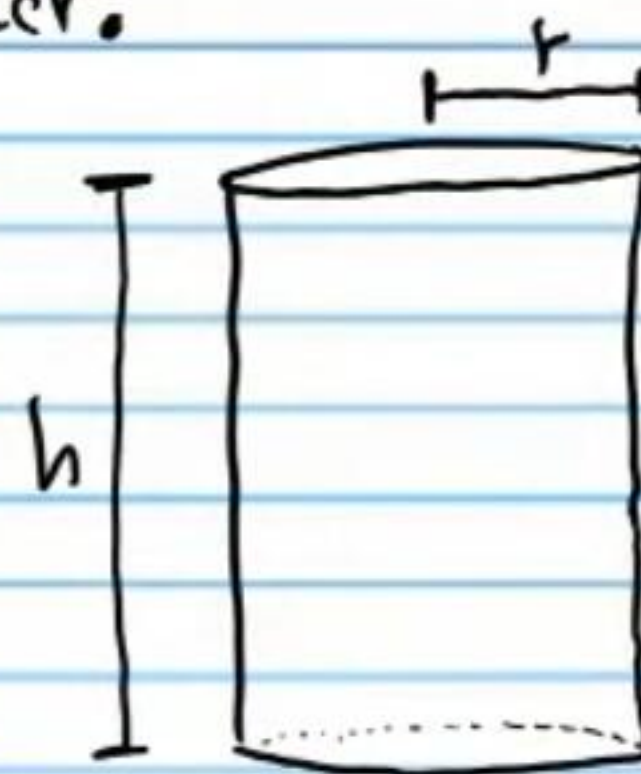
$$D = 2 \times r$$

37) Linda wants to make a fence around a circular garden. Approximately how many feet of fencing should she buy if the diameter of the garden is 87 feet?

38) If a crater has a radius of 3.6 Kilometers, approximately how many kilometers would a person have to walk to explore the edge?

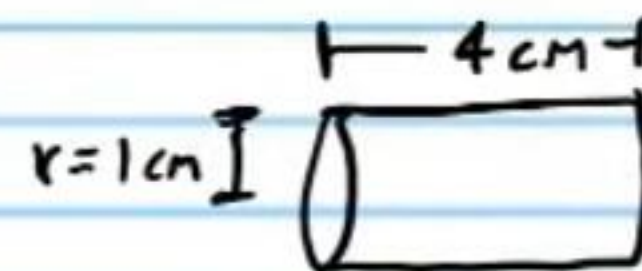
ix) Volume of a Cylinder:

$$V \approx 3.14 \times r^2 \times h$$



39) Find the approximate volume of a soda can in the shape of a cylinder with height 6 inches and a radius of three inches.

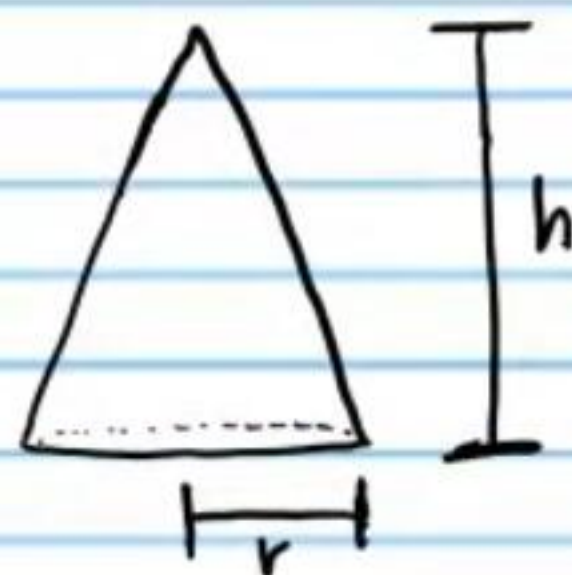
40) What is the approximate volume of a cake in the shape of a cylinder with length (i.e. height) four centimeters and a radius of one centimeter?



④ Approximate the volume of a cylinder with diameter eight inches and height 10 inches.

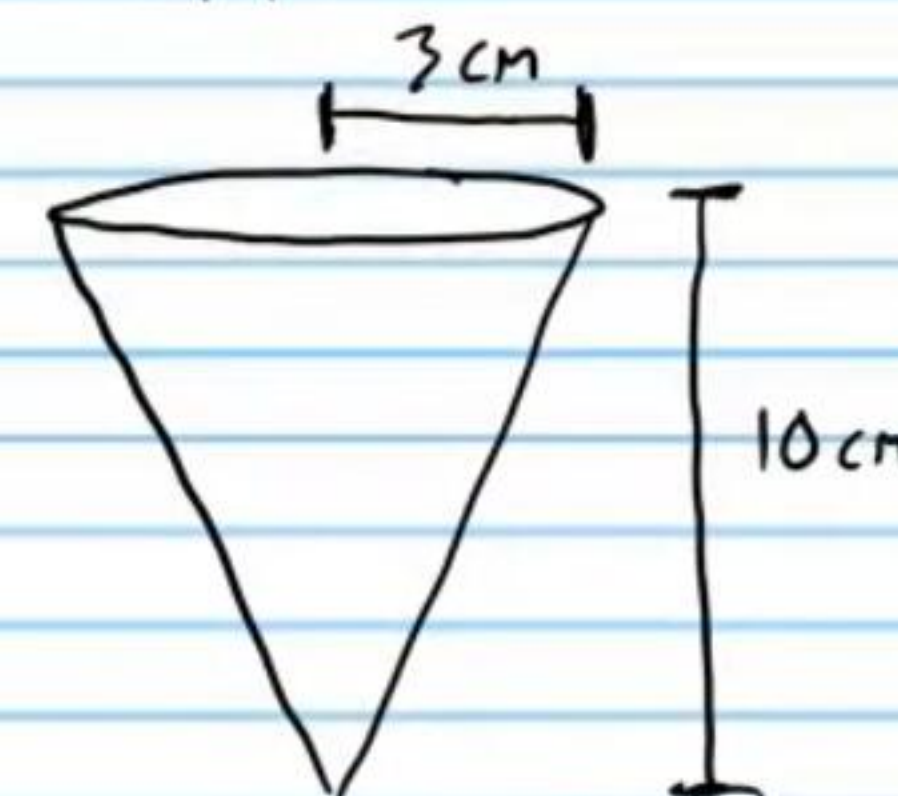
X) Volume of a Cone:

$$V \approx 3.14 \times r^2 \times h \div 3$$



④② What is the approximate volume of a pile of sand in the shape of a cone with a base radius of 2 feet and a height of 8 feet?

④③ Find the approximate volume of a snow cone with the dimensions below (a snow cone is a cone made out of paper with flavored ice in it).



④④ Approximate the volume of the cone below.

